# YANG, YAO

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## 1. SUMMARY

Completed **MSc in Artificial Intelligence at HKU** and **BEng in Data Science and Big Data Technology at NEU**, YANG, YAO combines a solid theoretical grounding in mathematics and statistics with deep practical experience in algorithm programming. Thus far, he has led **three first-author papers** (two accepted by ACL 2025 and ACM Multimedia 2025, one under review), and has been credited as **a core contributor on two technical reports**. As an early-stage researcher, his research interests focus on **(1) (Multimodal) Large Language Models**, **(2) Agents**, and **(3) Safe and Trustworthy AI**, while staying **OPEN to exploration across broader topics**. Currently, He is **pursuing PhD opportunities** for **Fall 2026** to further contribute to these domains.

## 2. EDUCATION

**MSc in Artificial Intelligence, University of Hong Kong** — **Sep 2023 - Jun 2025**
GPA of 3.24/4.00, Faculty of Science — Hong Kong SAR
► Core courses: Optimization Theory, Machine Learning, Computational Intelligence, and Deep Learning

**BEng in Data Science and Big Data Technology, Northeastern University** — **Sep 2019 - Jun 2023**
GPA of 3.76/5.00 (87.6/100.0), School of Mathematics and Statistics — Shenyang & Qinhuangdao
► Core courses: Advanced Algebra, Mathematical Analysis, Probability Theory, Bayes Statistics, Random Process, Python Programming, Distributed Clusters, MySQL Database, Operating System, and Computer Networks

## 3. PUBLICATIONS & SUBMISSIONS

1 additional paper not listed

[Note] One first-author work on agents and another first-author work on reinforcement learning are in progress. The acceptance decision for the papers under review will be released on April 30, AOE.

[1] (**ACL 2025**, **First Author**, **Poster**) A Mousetrap: Fooling Large Reasoning Models for Jailbreak with Chain of Iterative Chaos.

[2] (**ACM Multimedia 2025**, **First Author**, **Oral**) Argus Inspection: Do Multimodal Large Language Models Possess the Eye of Panoptes?

[3] (**Intern SafeWork Technical Report 2025**, **Core Contributor**) SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law.

[4] (**Under Review 2026**, **First Author**, **All Reviewers Recommend Acceptance**) Dr. Bench: A Multidimensional Evaluation for Deep Research Agents, from Answers to Reports.

[5] (**ACL 2026**, Co-author) SafeEvalAgent: Toward Agentic and Self-Evolving Safety Evaluation of LLMs.

[6] (**ACL 2026**, Co-author) Probing the Safety Robustness of LLMs in Latent Space.

[7] (**ICLR 2026**, Co-author, Poster) GhostEI-Bench: Do Mobile Agent Resilience to Environmental Injection in Dynamic On-Device Environments?

[8] (**OpenRT Technical Report 2026**, Co-author) OpenRT: An Open-Source Red Teaming Framework for Multimodal LLMs.

[9] (**AAAI 2026**, Co-author, Poster) The Other Mind: How Language Models Exhibit Human Temporal Cognition.

[10] (**AAAI 2026**, Co-author, Poster) CADiff: Context-Aware Diffusion for Controllable Anomaly Generation in Anomaly Detection.

[11] (**NeurIPS 2025**, Co-author, Poster) SafeVid: Toward Safety Aligned Video Large Multimodal Models.

[12] (Under Review 2026, Co-author, All Reviewers Recommend Acceptance) Towards Context-Invariant Safety Alignment for Large Language Models.

[13] (Under Review 2026, Co-author) Simulated Ensemble Attack: Transferring Jailbreaks Across Fine-tuned Vision-Language Models.

## 4. ACADEMIC EXPERIENCES

beyond the above publications and submissions

**Multimodal Value Alignment: Benchmark and Reward Model Fine-tuning** **Nov 2024 - Mar 2025**

► Constructed benchmarks with Multimodal Large Language Models (MLLMs, e.g., GPT, Gemini, and Qwen) based on the scenarios and elements of real-world human ethical values to evaluate the responses of MLLMs in terms of value alignment on response quality, security, reliability, and compliance.

► Built and optimized two pipelines based on image generation with DALL·E 3 and search with Google Image Search, and generated about 200,000 multimodal value data layer by layer through prompts engineering.

► Designed Outcome Reward Model (ORM) by full and Low-Rank Adaptation (LoRA) Supervised Fine-Tuning (SFT) Qwen2.5-VLs (7B, and 72B) and Process Reward Model (PRM), with accuracy rates of 93.9% and 96.8%.

**Stochastic Optimization Method for AUC Maximization** **Aug - Oct 2024**

► Explored the optimization methods for the Area Under receiver operating characteristic Curve (AUC) maximization based on the equivalent objectives, focusing on the fact that the classical stochastic optimization methods are not directly applicable to AUC maximization due to the nonlinearity of the objective functions.

► Designed diverse forms of alternative loss functions based on mathematical tricks (e.g., property of moment generating function, and Taylor expansion), and verified the convergence and stability of the algorithms.

► Achieved AUC scores of 0.9296 and 0.7197 respectively on the simple and difficult benchmarks of imbalanced classification tasks, surpassing the baseline.

## 5. INTERNSHIPS

**Shanghai Artificial Intelligence Laboratory, Safety and Trustworthiness AI Center** **Nov 2024 - Present**

Large Language Models Research Intern (Safety and Value) Shanghai

► Constructed multimodal value benchmarks using MLLM APIs based on image generation and image search, and trained 7B and 72B outcome reward models by full and LoRA SFT.

## 6. PROFESSIONAL SKILLS

**Python (PyTorch, TensorFlow, NumPy, Pandas, LLMs API, Matplotlib, Scikits-Learn)**, **R**, Julia, **MySQL**, Apache Hadoop Distributed Clusters (MapReduce & Zookeeper & Hive & HBase & Spark & Flume & Flink & Kafka), etc.

## 7. LEADERSHIP & TEAMWORK EXPERENCES

Student Assistant Principal, Northeastern University at Qinhuangdao Dec 2020 - Dec 2021

Coach of Debate Team, Sydney Smart Technology College, Northeastern University Sep 2020 - Feb 2021

## 8. QUALIFICATIONS & PRIZES & COMPETITIONS

Senior Big Data Application Engineer Certificate, Talent Training Project, Education and Examination Center, Ministry of Industry and Information Technology, P.R.China Mar 2022

University Scholarships (4 times) & Innovation and Entrepreneurship Scholarships (2 times) Undergraduate

Honourable Mention, Mathematical and Interdisciplinary Contest in Modeling (MCM-ICM) May 2022

Third Prize, Chinese Mathematics Competitions (CMC) Dec 2020